Calibration-free P300 decoding

Cross-subject transfer on BrainInvaders2013a

Birzhan Zhunusbekov Arman Abakov
Nazerke Khamzina Bekmurat Amangeldiyev

17 subjects Linear ERP decoder Offline evaluation

Research question and hypotheses

Can other users' data replace a new user's calibration?

H1 LOSO ranks targets above chance

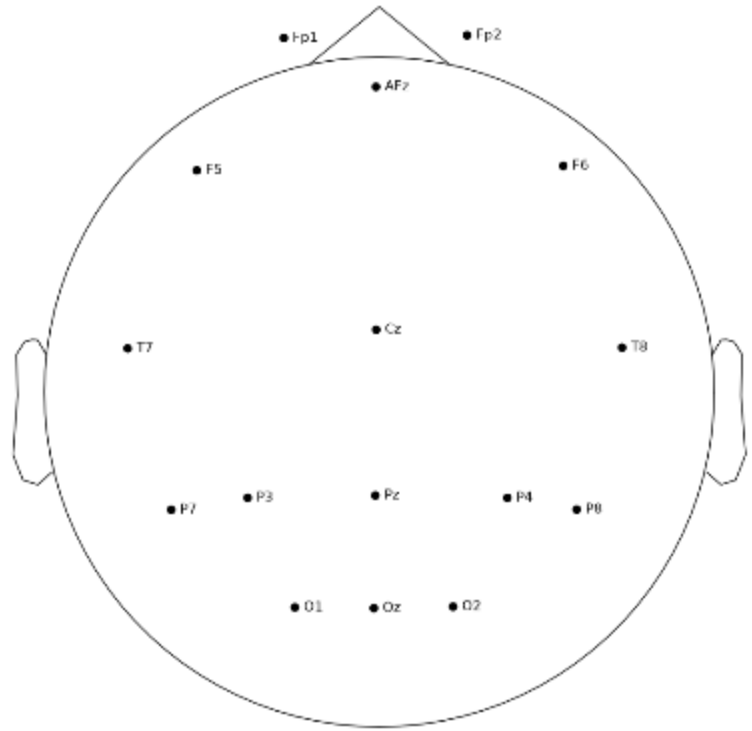
H2 Within-subject training exceeds LOSO

H3 The gap remains with matched training counts

Earlier subjects 8–10 informed development. We disclose that boundary.

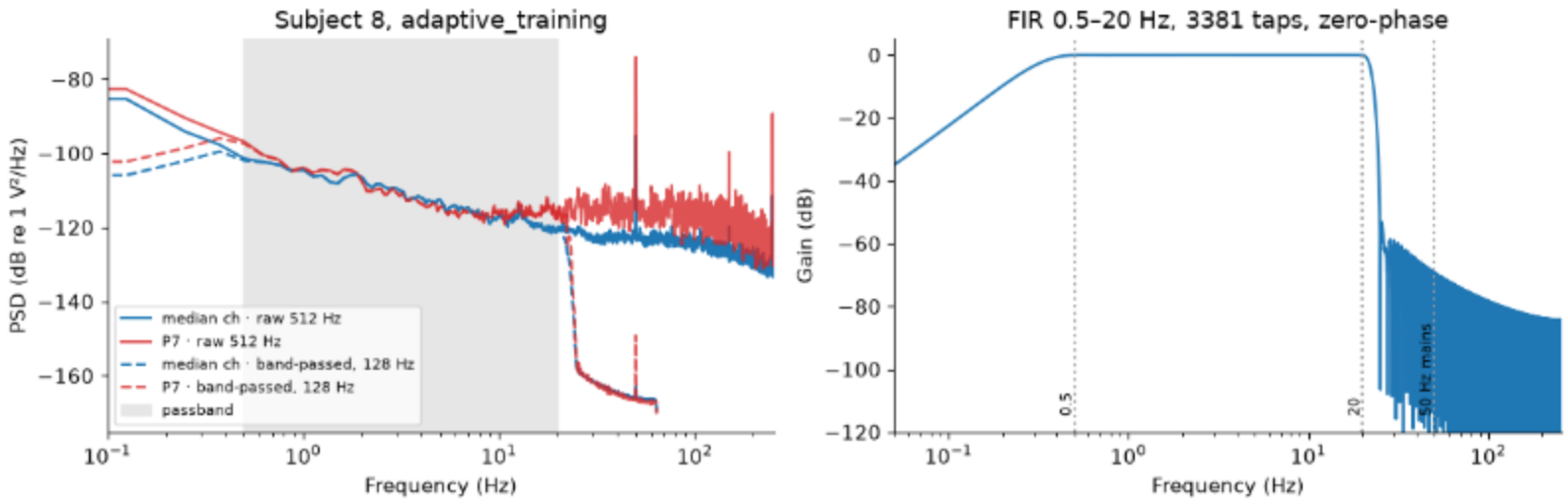
Data and provenance

Property	Audited value
Subjects	8–24, n = 17
Runs	2 training runs per subject
Acquisition	16 EEG channels, 512 Hz
Reference	Left earlobe
Events	80 targets / 400 non-targets per run



Original Zenodo training files. MOABB creates the MNE objects.

Filtering and resampling



FIR 0.5–20 Hz

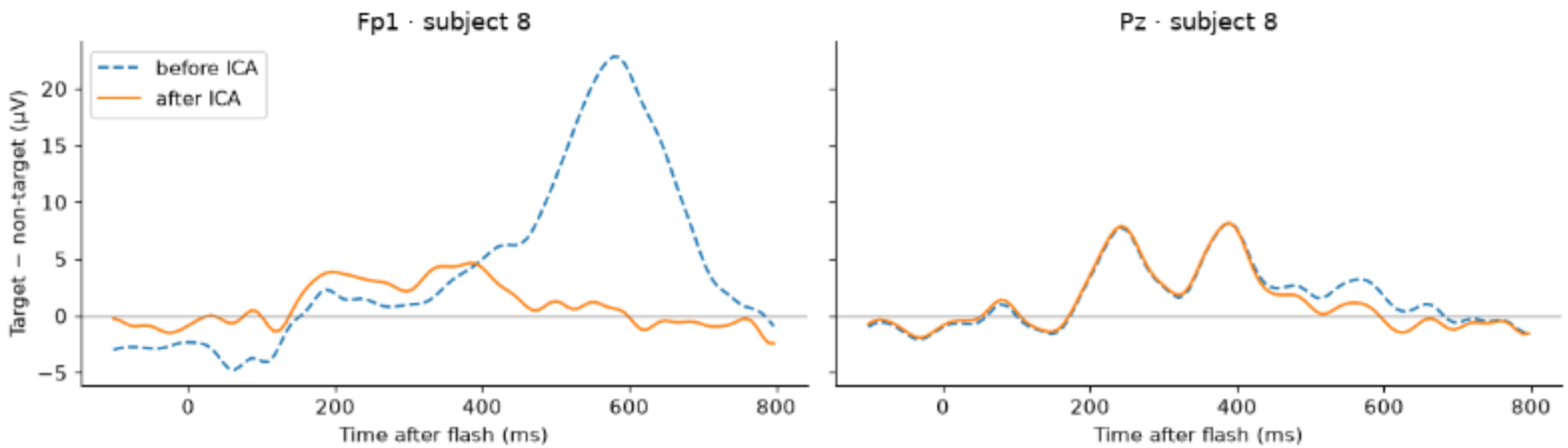
512 to 128 Hz

Events resampled together

Recording-edge guard

Per-run processing preserves run boundaries

Artifact handling and quality control



97.40% retained

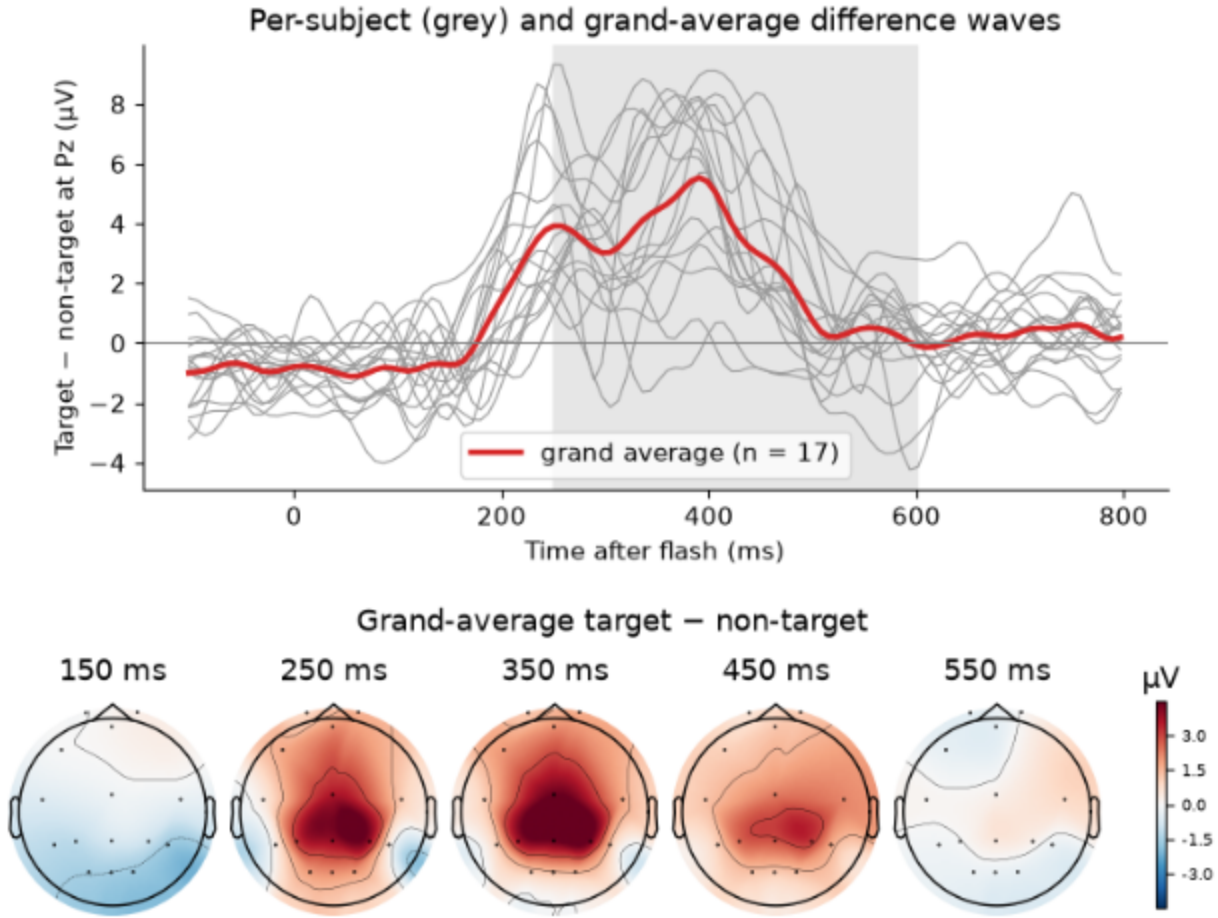
Edge drops
3 target / 12 non-target

Amplitude drops
80 target / 329 non-target

100 µV peak-to-peak

ICA is offline and label-free. Residual artifacts remain possible.

P300 physiology before decoding



Fixed Pz window 250–600 ms

16/17 positive subject window means

Mean contrast 2.74 µV

Features and linear classifier

Step	Fixed specification
Channels	Cz, P3, Pz, P4, Oz
ERP features	16 × 50 ms windows over 0–800 ms
Dimension	5 × 16 = 80 features
Scaling	Training-fold StandardScaler
Classifier	Shrinkage LDA, analytic shrinkage

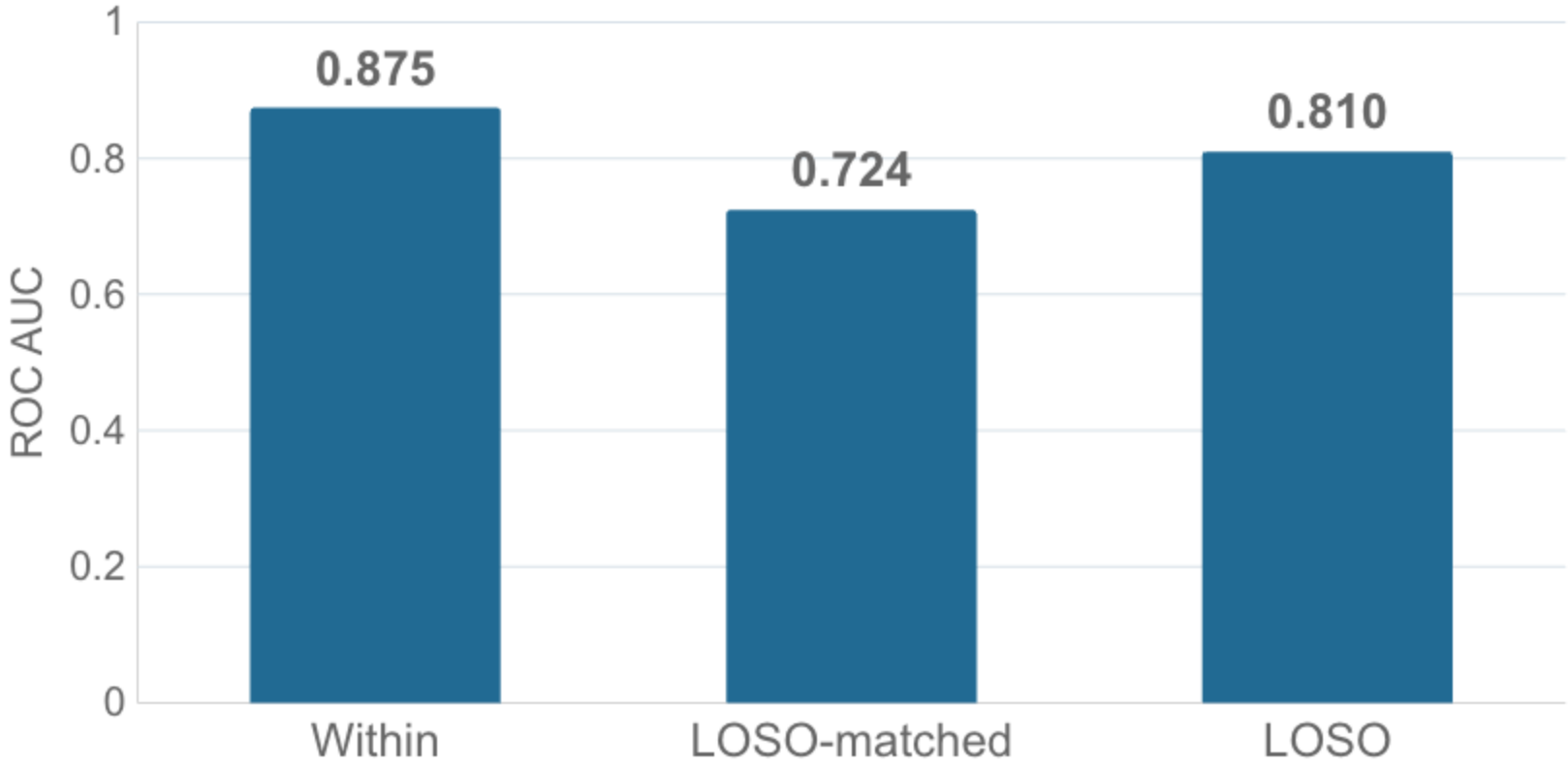
No pooled bad-channel flags shape training features

Evaluation on the same held-out epochs

Condition	Training data	Test data
Within	Subject's other run	Same held-out run
LOSO-matched	Other users, matched class counts	Same held-out run
LOSO	All runs of the other 16 users	Same held-out run

Run groups for Within. Entire subjects held out for transfer.
Chance AUC = 0.5. One mean over two test runs per subject.

Mean AUC across subjects



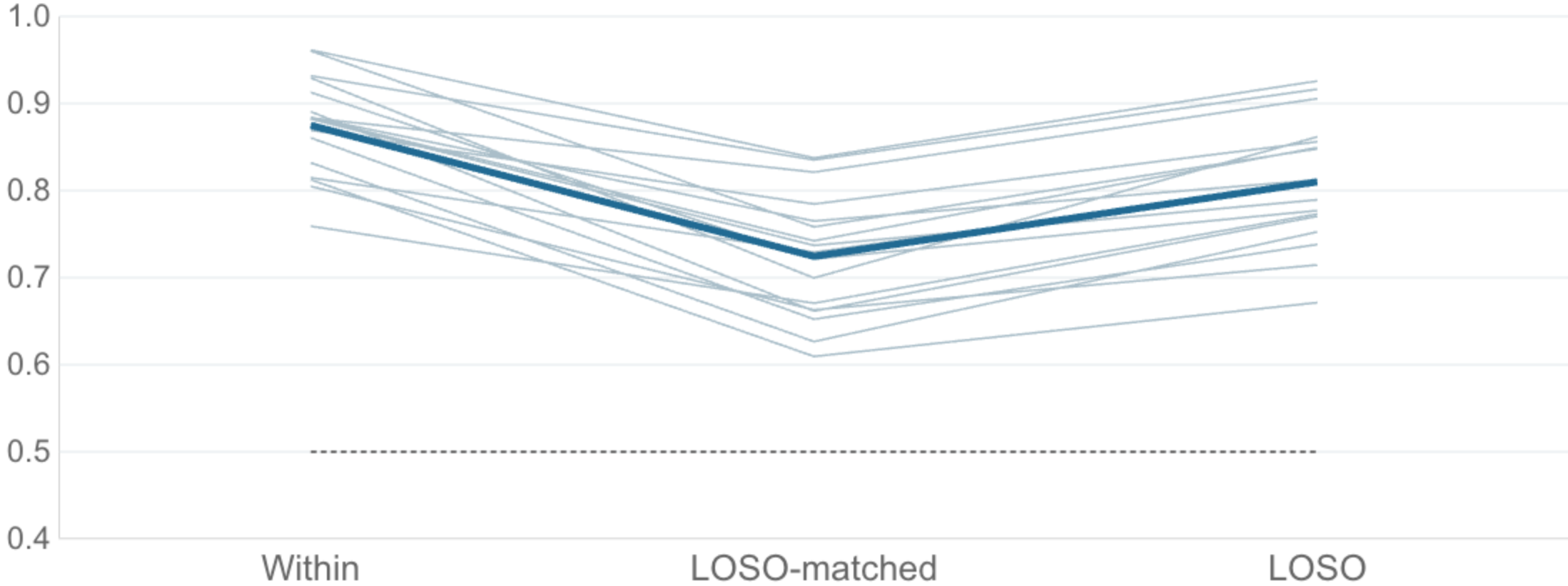
Mean ± SD

Within
0.875 ± 0.056

Matched
0.724 ± 0.071

LOSO
0.810 ± 0.072

Individual transfer performance



Median Within–LOSO gap: 0.072 Median Within fold SD: 0.018

Paired hypothesis tests

Hypothesis	Median gap	Holm p
H1 LOSO > 0.5	0.309	2.29e-5
H2 Within > LOSO	0.072	1.91e-4
H3 Within > LOSO-matched	0.142	2.29e-5

Pilot-excluded sensitivity: n = 14
Holm p for H1 / H2 / H3: 1.83e-4 / 3.05e-4 / 1.83e-4

Conclusion and limits

Supported in this cohort: H1, H2, H3

Transfer performance must be assessed per user

Offline cleaning uses future unlabelled samples

Single-flash AUC does not establish online command accuracy

Independent cohorts are needed for stronger generalization claims

References and AI disclosure

Vaineau et al. (2019). Brain Invaders dataset. arXiv:1904.09111

Barachant & Congedo (2014). Plug&Play P300. arXiv:1409.0107

Kim & Kim (2025). Zero-training P300. DOI: 10.1109/TNSRE.2025.3603979

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AI assistance: OpenAI Codex supported implementation, execution, checks and drafting. The team reviews and defends the submitted work.

Reproducibility record

Package	Version
mne	1.13.2
moabb	1.7.2
numpy	2.5.4
scipy	1.18.1
pandas	2.3.3
scikit-learn	1.9.1
matplotlib	3.11.2
PyYAML	6.0.3
remotezip	0.12.6
threadpoolctl	3.7.0

17 subjects
Seed 613

Full run: 7.41 min
Google Colab

Empty data cache
Verified